

The empirical recurrence for OEIS A300172, recovered from its tail

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Abstract

OEIS A300172 records “Empirical recurrence of order 66” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 103$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 70$. Its last coefficient is $c_{66} = 1168 \neq 0$, so no further tail satisfies anything shorter; any order-66 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A300172 is “Number of $n \times 5$ 0..1 arrays with every element equal to 2, 3, 4, 5, 6, 7 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

0, 17, 192, 2620, 50419, 886940, 16262433, 297460457, ...

The entry states:

Empirical recurrence of order 66 (see link above)

As of the “Last modified” line on the live entry (Feb 27 2018 12:17:41, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 103$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 103$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 66: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 66.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 206$ exact terms from $n = 4$ on, it returns order 66 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{66} c_i a(n-i),$$

c_1	18	c_{18}	−6750122629	c_{35}	206410676083	c_{52}	15555425365
c_2	90	c_{19}	4178613449	c_{36}	−1549740137864	c_{53}	13526045387
c_3	−1291	c_{20}	4877599264	c_{37}	1096593370085	c_{54}	−16051288204
c_4	−6497	c_{21}	−12578128784	c_{38}	585991716938	c_{55}	−5484146920
c_5	30572	c_{22}	68061810227	c_{39}	−113989126690	c_{56}	−3745417034
c_6	174107	c_{23}	−43028003619	c_{40}	−205182392665	c_{57}	2484447210
c_7	−232436	c_{24}	63763115016	c_{41}	−80129337520	c_{58}	2813948354
c_8	−1345841	c_{25}	−70580012542	c_{42}	−424306324125	c_{59}	606869589
c_9	−647282	c_{26}	−70912982081	c_{43}	−139444435009	c_{60}	−98847854
c_{10}	−4801011	c_{27}	80147687579	c_{44}	289591039891	c_{61}	−63747136
c_{11}	10324369	c_{28}	−596535804504	c_{45}	458121549660	c_{62}	−16794542
c_{12}	30101157	c_{29}	941067077629	c_{46}	−247059602071	c_{63}	−2239380
c_{13}	123534112	c_{30}	−482007647604	c_{47}	−7116148451	c_{64}	151820
c_{14}	−21507313	c_{31}	−112956955308	c_{48}	−291708291936	c_{65}	50400
c_{15}	53473930	c_{32}	1737847655222	c_{49}	172285949498	c_{66}	1168
c_{16}	−1190021419	c_{33}	−1870368176003	c_{50}	193626950996		
c_{17}	410112955	c_{34}	680998750295	c_{51}	−146236282561		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 70$, and it is the recurrence OEIS A300172 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{66} - c_1 x^{65} - \dots - c_{66}$. Its constant term is $-c_{66} = -1168$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$

vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 66 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 66, so $\deg q = 66$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 66, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 66, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1168.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-66 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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